

AI and Robotics

Mitsubishi to mass-produce 1,000 humanoid robots monthly at unused engine line

HL Human, unveiled in 2025, features 19 DOF and AI-powered autonomy for complex industrial tasks and human-like movement.

READ NEXT: Aurora project moves toward 47-hole drilling amid renewed US nuclear fuel push

By Jijo Malayil Aug 05, 2026 07:55 AM EST



Highlanders' Humanoid robot. Representative image.

Highlanders

Automaker Mitsubishi will begin mass-producing artificial intelligence-powered humanoid robots at its Kyoto plant in Japan.

The robots, developed by Tokyo-based startup Highlanders Inc., will be built on an unused engine production line repurposed for robot assembly.

The facility is expected to produce up to 1,000 humanoid robots each month for industrial use.

The move reflects growing investment by automakers in physical AI and factory automation as manufacturers seek to expand the deployment of humanoid robots in production environments.

00:00

00:00

00:24

Factory robots rise

Mitsubishi has partnered with Tokyo-based robotics startup Highlanders to develop and mass-produce AI-powered humanoid robots, with production scheduled to begin in early 2027 at the company's Kyoto Plant.

The facility is expected to manufacture up to 1,000 humanoid robots per month by repurposing unused factory space and idle engine production lines. The first robots will be deployed on Mitsubishi's engine manufacturing lines, where they will perform factory tasks and generate operational data to refine their performance. After this validation phase, Mitsubishi and Highlanders plan to

market the humanoid robots to other manufacturers and businesses facing persistent labor shortages.

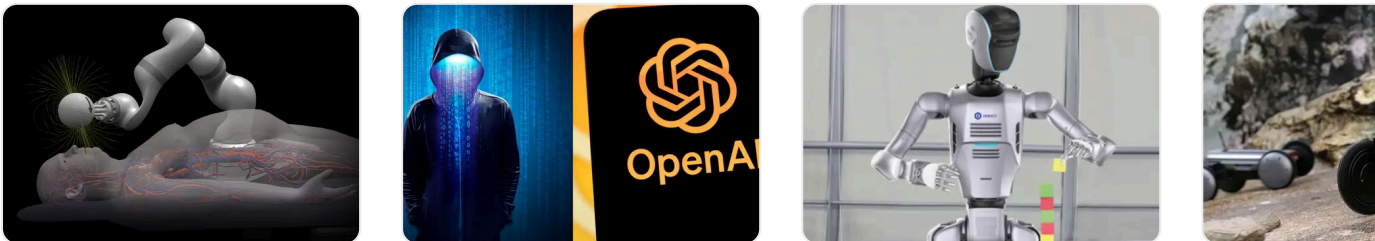


Mitsubishi Motors has already invested in Highlanders and plans to make additional investments in the future.

The partnership combines Highlanders’ physical [AI](#) technology with Mitsubishi’s decades of automotive manufacturing expertise and quality control processes. By leveraging its established production capabilities, Mitsubishi aims to scale humanoid robot manufacturing efficiently while ensuring automotive-grade reliability.

MORE FROM AI AND ROBOTICS

[See All](#)



US Backs Autonomous Stroke Treatment Robot From MIT Spinout With \$32 Million...

OpenAI Locks Down Astra After Model Raises First-Ever Critical Cyber Capability Fears

'World's First Embodied All-Terrain' Humanoid For Life Beyond The Factory Unveiled

World's First Au That Drives Abo Tested By Hyun

“By utilizing humanoid [robots](#) in our own manufacturing facilities and supporting the production of Highlanders products, we aim to leverage the outcomes of this collaboration to drive our growth and enhance corporate value,” said Takao Kato, chairman of the board and CEO of Mitsubishi Motors Corporation, in a [statement](#).

Tokyo startup advances

Highlanders is a University of Tokyo spin-off founded in 2023 with the goal of developing general-purpose humanoid and quadruped robots powered by physical AI. Headquartered in Tokyo, the startup focuses on building robots capable of performing practical work in manufacturing, logistics, infrastructure inspection, and disaster response, where labor shortages are becoming increasingly severe.

The company’s flagship humanoid platform, HL Human, was first unveiled in 2025. The prototype features a 19-degree-of-freedom (DOF) body equipped with high-output electric actuators that enable human-like movement. It is designed to autonomously perform complex industrial tasks using AI-powered perception, motion planning, and natural-language interaction.

Highlanders has also announced plans to equip future versions with dexterous five-finger hands capable of grasping objects weighing up to 3 kilograms,

significantly expanding the robot's manipulation capabilities.

The startup's broader vision extends beyond hardware. Highlanders is developing a physical AI framework that allows robots to understand instructions, perceive their surroundings, and execute tasks with minimal human intervention. The company also provides robot simulation software to accelerate training and deployment across different industries.

“We believe that achieving mass production of domestically developed humanoid robots through this partnership with [Mitsubishi](#) Motors, which brings decades of manufacturing expertise, represents a significant step toward achieving this goal. Furthermore, deploying our robotics technologies within Mitsubishi Motors' manufacturing operations will provide a valuable opportunity to drive technological advancement in our products,” said Hiroya Masuoka, representative director and CEO of Highlanders, in a statement.

RECOMMENDED ARTICLES

[Men Over 40 Are Ditching the Blue Pill for This 7x Stronger Alternative](#)





Get the latest in engineering, tech, space & science - delivered daily to your inbox.

Enter Your Email

Sign up for free

By subscribing, you agree to our [Terms of Use](#) and [Policies](#)
You may unsubscribe at any time.

0 COMMENT >

Subscribe to **IE+** Today!

Access to exclusive content, expert insights and a deeper dive into engineering and tech. No ads, no limits.

Explore Now!

By **Jijo Malayil**

Jijo is an automotive and business journalist based in India. Armed with a BA in History (Honors) from St. Stephen's College, Delhi University, and a PG diploma in Journalism from the Indian Institute of Mass Communication, Delhi, he has worked for news agencies, national newspapers, and automotive magazines. In his spare time, he likes to go off-roading, engage in political discourse, travel, and teach languages.

TRENDING

LATEST

- 1 **Aurora project moves toward 47-hole drilling amid renewed US nuclear fuel push** →
- 2 **China expands world's largest rare earth mine by 50% to 15 million tonnes annually** →

- 3 **Trump reveals \$2B investment plan to boost US domestic critical mineral supply chain** →
- 4 **US Air Force funds new goggles to shield pilots from blinding nuclear blast flash** →
- 5 **SpaceX promises self-sustaining power and water recycling for \$55b Grimes County project** →

Men Over 40 Are Ditching the Blue Pill for This 7x Stronger Alternative



Sponsored by trending-hunt.com

US can destroy all
Russia, China nuclear...
sites in just 2 hrs: Study



Interesting Engineering

Round 2: Elon Musk
now says he will fight...
Putin using only his left
hand



Interesting Engineering

FEATURED STORIES



Nobody told me that Androids could do this! Say goodbye to ads forever!

[Experts On Tech](#)



Meet Mature Women 50+ Online

Site where feeling shy is normal.

alofhearts.com



Stop Scrolling. Mature Women Are Online Here.

Site where feeling shy is normal.

goldenromances.com

Follow Us On

Energy

Science

Military

Health

Transportation

Space

Innovation

Culture

News

Videos

Podcasts

Photo Stories

Webinars

Engineers Directory

IE+ Premium

Columns

Case Studies

Research

Interviews

Newsletters

IE Academy

IE Awards

About Us

Advertise

Contact Us

Subscription

Policies

Site Map



[Privacy Policy](#) | [AI and Ethics](#) | [Editorial Policy](#) | [Terms of Services](#)

[Do Not Sell My Personal Information](#) | [Privacy Manager](#)

© Copyright 2025 | IE Media, Inc. | All Rights Reserved